

ONLINE EXAM PORTAL

Use Case Flow

This document describes the goal, flow, and outcome of the primary use cases identified for the Online Exam Portal, corresponding to the Use Case Diagram.

UC1: Login	
Actor(s)	Student, Teacher, Administrator
Goal	To authenticate the user and grant role-based access to the portal.
Precondition	The user has a registered account on the portal.
Main Flow	1. The user opens the Online Exam Portal login page. 2. The user enters their registered username/email and password. 3. The system validates the credentials. 4. The system grants access based on the user's role (student, teacher, or administrator).
Alternate Flow	- If the credentials are invalid, the system displays an error and allows the user to retry. - If the user has forgotten their password, they may use a password-reset option.
Postcondition	The user is logged in and directed to their respective dashboard.

UC2: View Exam Schedule	
Actor(s)	Student
Goal	To let the student view upcoming exams and plan accordingly.
Precondition	The student is logged in.
Main Flow	1. The student navigates to the exams section. 2. The system displays available exams with subject, date, duration, and number of questions. 3. The student selects an exam to view further details.
Alternate Flow	- If no exams are currently scheduled, the system displays an appropriate message.
Postcondition	The student is aware of upcoming exams and their schedules.

UC3: Attend Exam	
Actor(s)	Student
Goal	To allow the student to complete an examination within the allotted time.
Precondition	The exam's scheduled start time has arrived and the student is logged in.

Main Flow	1. The student starts the exam within the scheduled time window. 2. The system displays questions along with a countdown timer. 3. The student answers questions and navigates between them, marking any for review. 4. The system automatically saves answers whenever a response is selected or changed (include: Auto-Save Answers). 5. The student submits the exam, or the system auto-submits when the timer reaches zero (include: Auto-Submit Exam).
Alternate Flow	- If the student loses network connectivity temporarily, previously saved answers are retained and the exam resumes from the last saved state. - If the student attempts to access the exam before the scheduled start time, the system denies entry.
Postcondition	The student's responses are recorded and the exam is marked as submitted.

UC4: Auto-Submit Exam and Automatic Evaluation

Actor(s)	System (triggered during Attend Exam)
Goal	To close the exam automatically and compute the student's score without manual intervention.
Precondition	An exam is in progress for a student.
Main Flow	1. The countdown timer reaches zero, or the student manually submits the exam. 2. The system locks further changes to the student's responses. 3. The system automatically evaluates objective-type questions (include: Automatic Evaluation). 4. The system stores the responses and computed results in the database.
Alternate Flow	- If the exam contains only subjective questions, the system marks it as pending manual evaluation instead of computing a score.
Postcondition	The exam is submitted and scored, and results are available for the student to view.

UC5: View Results

Actor(s)	Student
Goal	To let the student check their marks and performance after an exam.
Precondition	The exam has been evaluated.
Main Flow	1. The student navigates to the results section. 2. The system retrieves and displays the student's marks and performance summary.
Alternate Flow	- If results are not yet published, the system informs the student that evaluation is in progress.
Postcondition	The student is informed of their exam outcome.

UC6: Manage Question Bank

Actor(s)	Teacher
Goal	To maintain an up-to-date set of questions for use in examinations.
Precondition	The teacher is logged in with appropriate access privileges.
Main Flow	1. The teacher navigates to the question bank. 2. The teacher adds, edits, or deletes questions, including multiple-choice options. 3. The system saves the updated question bank for use in examinations.
Alternate Flow	- If a question is already in use in a scheduled exam, the system warns the teacher before allowing deletion.
Postcondition	The question bank is up to date and ready for exam creation.

UC7: Create / Manage Exam

Actor(s)	Teacher, Administrator
Goal	To configure and schedule examinations for students.
Precondition	The user is logged in with teacher or administrator privileges.
Main Flow	1. The user selects the option to create or edit an exam. 2. The user specifies the exam title, duration, marks, schedule, and associated questions. 3. The system validates and saves the exam configuration.
Alternate Flow	- If required fields are missing or invalid, the system prompts the user to correct them before saving.
Postcondition	The exam is created or updated and made available to students as per its schedule.

UC8: Generate Reports

Actor(s)	Administrator
Goal	To provide a consolidated view of student performance and examination statistics.
Precondition	One or more exams have been completed and evaluated.
Main Flow	1. The administrator navigates to the reports section. 2. The administrator selects an exam or a reporting period. 3. The system compiles student performance and examination statistics. 4. The system displays or exports the generated report.
Alternate Flow	- If no data is available for the selected exam or period, the system displays an appropriate message.
Postcondition	The administrator obtains a report summarizing exam performance and statistics.